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VIP Glycoconjugates Very Important Paper

Klebsiella pneumoniae Glycoconjugate Vaccine Leads Based on Semi-Synthetic O1 and O2ac Antigens

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Abstract: *Klebsiella pneumoniae* (KP) is a common opportunistic pathogen that emerged as a new critical threat to human health, due to its hypervirulence and widespread resistance against many antibiotics, including carbapenems. Alternative intervention strategies such as vaccines are not available. Cell-surface lipopolysaccharides (LPS) and capsular polysaccharides (CPS) are attractive targets for vaccine development. We present a method to synthesize LPS substructures, covering over 70 % of virulent KP strains that were used to characterize the antibody repertoire of infected patients. Thereby, glycoconjugate vaccine leads against the *Klebsiella pneumoniae* serotypes O1, O2a, O2afg and O2ac have been identified.

Introduction

Klebsiella pneumoniae (KP), a Gram-negative bacterium of the Enterobacteriaceae family,[1] can cause serious infections[1–2] with major clinical implications[3] and is responsible for a high case fatality rate of up to 50 % in infected patients.[4] Multidrug resistance,[5] in particular against carbapenem antibiotics,[6] earns KP a spot on the WHO critical priority pathogen list.[7] Mortality is particularly high in patients infected with carbapenemase-positive KP,[8] hence there is an urgent need for alternatives to antibiotics.

Cell-surface carbohydrates are crucial for the viability and pathogenicity of bacteria such as KP. Different capsular polysaccharides (CPS), K-antigens and lipopolysaccharide (LPS) O-antigens form the bacterial capsule and are used to distinguish more than 80 KP serotypes.[9] LPS, mainly involved in resistance to complement bactericidal activity,[10] consists of lipid A, a core oligosaccharide and an O-antigenic polysaccharide (O-PS). The O-PS plays a key role in host-pathogen interactions.[11]

Most clinical isolates carry O-antigens that are composed of either galactose (O1 and O2 antigens) or mannose (O3 and O5 antigens).[9b,c] KP serotype O1 antigen consists of two types of polysaccharides: (1) O2a with a [→3)-*β*-D-Gal*f*-(1→3)-*α*-D-Gal*p*-(1→] linked to O1 antigen repeating unit composed of a [→3)-*β*-D-Gal*p*-(1→3)-*α*-D-Gal*p*-(3→] and (2) O2afg antigen consisting of a core trisaccharide repeating unit composed of a [→3)-*β*-D-Gal*f*-(1→3)[*β*-D-Gal*p*-(1→4)]-*α*-D-Gal*p*-(1→] linked to O1 antigen[12]. KP serotype O2ac antigen consists of a tetrasaccharide repeating unit [→5)-*β*-D-Gal*f*-(1→3)-*β*-D-GlcNAc-(1→5)-*β*-D-Gal*f*-(1→3)-*β*-D-Gal*p*-[1] and is extended from the O2a antigen composed of

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galactopyranose (Gal*p*) and galactofuranose (Gal*f*) residues (Figure 1).

Novel vaccine candidates are urgently needed as no vaccines are available to prevent KP infections.[13] O-PS induces an immune response in the infected host[13b,14] and passive immunization with O-PS antibodies is protective in animal models.[13b,15] Conjugates of purified O-antigen polysaccharides and carrier proteins tetanus toxoid or *Pseudomonas aeruginosa* flagellin A and flagellin B have shown some promise as active immunization strategy against drug-resistant strains of KP in animal models.[15b,16]

A vaccine candidate composed of purified O1 and O2 LPS showed short-lived protection in mice.[16b] Antibody titers against O1 antigens were found to be stronger than against O2a antigens in three patients infected with KP O1.[17] However, protective epitopes of O1, O2a, O2afg and O2ac antigens remain unknown. Despite promising immunological results, the development of LPS-based vaccines was abandoned due to the challenges associated with LPS purification. Hot phenol extraction of LPS is time-consuming, expensive and results in toxic lipid A impurities.[18]

Synthetic, well-defined glycan epitopes of defined length and sequence can induce robust and specific immune responses as a promising alternative to native LPS extraction and purification.[19] However, the chemical synthesis of oligosaccharides resembling the KP O1 and O2ac antigens poses multiple challenges. The synthesis of the KP O1 antigen requires the installation of 1,2-*cis* linkages, has to overcome the poor nucleophilicity of the axial C4-hydroxyl of galactose and needs to furnish densely functionalized glycans (Figure 1A).

Here, we describe the syntheses of oligosaccharides resembling the LPS O1-O2a, O1-O2afg and O2c-O2a (O2ac) antigens as tools to facilitate in-depth immunological studies into the immunogenicity and potential epitope structure of the LPS. Sera of patients, infected with KP and carbapenem-resistant *Klebsiella pneumoniae* (CRKP), were screened on glycan arrays to identify leads for the development of glycoconjugate vaccines.

Results and Discussion

The O-antigenic polysaccharides in KP serotypes O1 and O2ac are polymers consisting of O1-O2a, O1-2afg, and O2c-O2a antigens, respectively (Figure 1A). Retrosynthetic analysis suggests that oligosaccharides **1**, **2** and **6** can be obtained from protected glycans **3**, **4** and **7** via functional group interconversions (FGI) and global deprotection. A linear glycosylation approach relies in turn on building blocks **8**–**14** (Figure 1B and 1C). Highly functionalized, branched heptasaccharide **3** presents a significant synthetic challenge. Fully protected heptasaccharide **3** can be obtained by coupling tetrasaccharide acceptor **5**, with galactose **11**. The poor nucleophilicity of the C4 hydroxyl group of **5** contributes to the low-yielding coupling (Figure 1B). A retrosynthetic

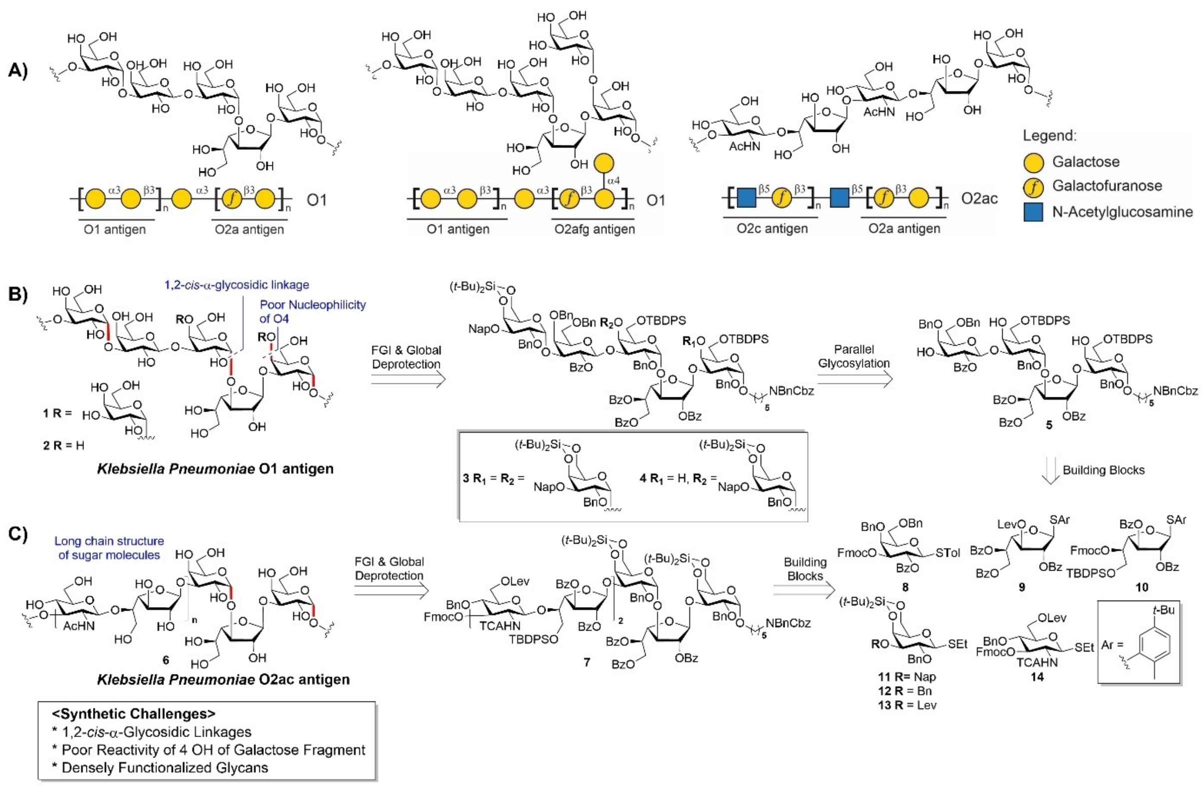


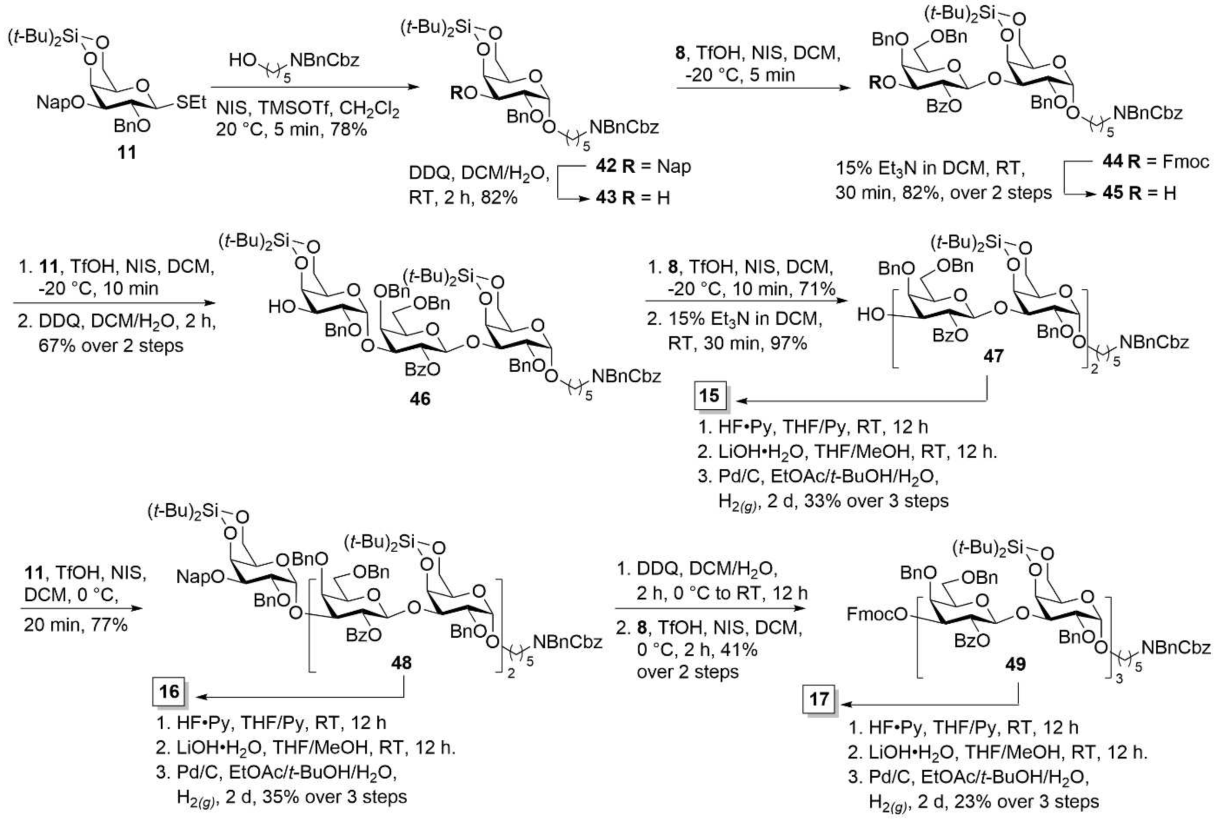
Figure 1. KP lipopolysaccharides A) Repeating units of KP O1 and O2ac serogroup antigens; Retrosynthetic analysis of B) KP O1 and C) KP O2ac antigens. Ac = acetyl, Bn = benzyl, Bz = benzoyl, 2-NAP = 2-naphthylmethyl, Lev = levulinyl, Fmoc = 9-fluorenylmethyloxycarbonyl, Cbz = carboxybenzyl, TBDPS = *tert*-butyldiphenylsilyl, (*t*-Bu)$_2$Si = *tert*-dibutylsilyl and TCA = Trichloroacetyl.

analysis of oligosaccharide **6** resembling the O2ac antigen presents several glycosylation challenges en route to the heptasaccharide. Electron-withdrawing groups reduce the nucleophilicity of glycosyl acceptors, in the context of densely functionalized glycans. Stereoselectivity for the 1,2-*cis*-*α*-glycosidic linkage of galactopyranosides **11–13** is ensured by the 4,6-*O*-di-*tert*-butylsilylidene (DTBS) group, while the 2-*O*-benzoyl (OBz) group on galactofuranosides **9**, **10** and tricholoacetamide (TCA) group on glucopyranoside **14** provide *β*-selectivity (Figure 1C).

In order to identify key glycan epitopes, 27 oligosaccharides (**15–41**) were designed to aid detailed immunological studies (Figure 2A and B). Oligosaccharides **15–17** resemble key epitopes of O1 antigens, while **18–20** mirror O2a antigens. Disaccharide **32** and trisaccharide **33**, containing a galactofuranose (Gal*f*) backbone with a galactose (Gal) branch, represent the O2afg antigen. The entire length of KP O2a-O1 (oligosaccharides **21–31**) and O2afg-O1 antigens **34** and **35** are covered to investigate the role of highly branched sequences in antibody binding (Figure 2A). Synthetic glycans related to the O2ac antigen (Figure 2B) included shorter fragments: monosaccharides **36–37**, trisaccharide **38** and tetrasaccharide **39** constitute the backbone of the O2c antigen. Oligosaccharides **40** and **41** correspond to one repeating unit of the O2ac antigen. All synthetic oligosaccharides are equipped with an aminopentyl linker at the reducing end for microarray studies and conjugation to the carrier protein for immunological studies.

The synthesis of building blocks **8**, **11–14** relied on modified literature procedures,[19a] while the synthesis of galactofuranosides **9** and **10** from commercially available D-galactose was newly developed (Supporting Information, Scheme S1).[20] Oligosaccharides **15–17**, resembling the O1 antigen, bear an *α*-galactose aglycone at the reducing end and were obtained via linear glycosylation using building blocks **8** and **11**. The reaction of donor **11** with a protected aminopentanol linker (*N*-(benzyl)benzyloxycarbonylaminopentanol) in the presence of trimethylsilyl trifluoromethanesulfonate (TMSOTf) and *N*-iodosuccinimide (NIS) gave **42** in 78 % with complete *α*-selectivity ($^1J_{C\text{-}H} = 170.1$ Hz; Supporting Information).

Oxidative cleavage of the naphthylmethyl ether (NAP) protecting group by 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) in DCM/H_2O (7/1) provided acceptor **43** in 82 % yield (Scheme 1). Donor **8** was reacted with 3-hydroxyl of acceptor **43** using NIS/triflic acid (TfOH) to obtain **44**, before Fmoc cleavage using 15 % Et_3N in DCM afforded compound **45** in 82 % yield over two steps with complete *β*-selectivity. Coupling of glycosyl donor **11** with acceptor **45** followed by the cleavage of the Nap group gave trisaccharide acceptor **46** in 67 % yield over two steps. Linking trisaccharide acceptor **46** and donor **8** followed by the removal of the Fmoc ester group, generated tetrasaccharide **47** as a single isomer. Pentasaccharide **48** was obtained by stereoselective *α*-galactosylation ($^1J_{C\text{-}H} = 171.4$ Hz) of thioglycoside donor **11** and acceptor **47**. Removal of the NAP ether in **48** followed



Scheme 1. Synthesis of O1 antigen-related oligosaccharides **15–17**,

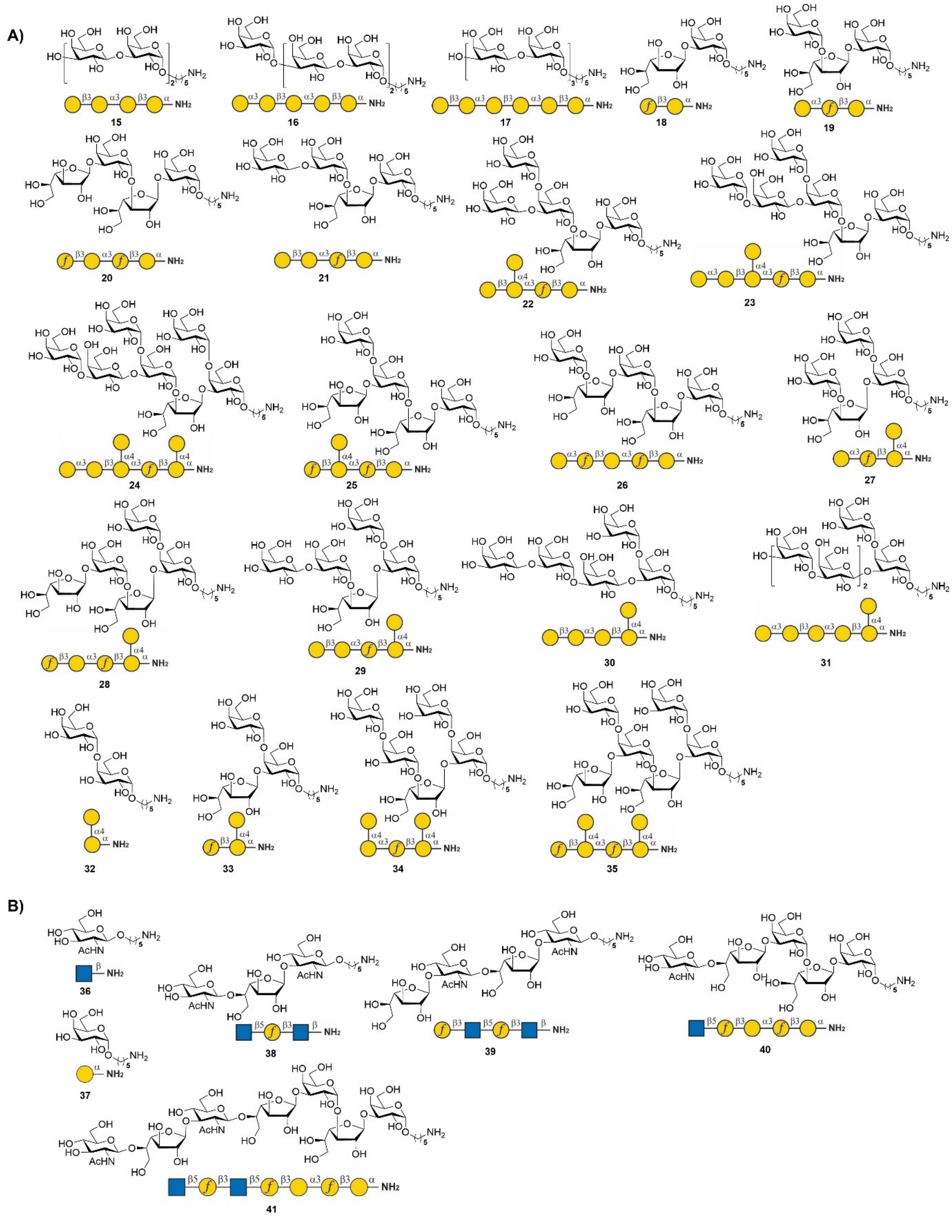
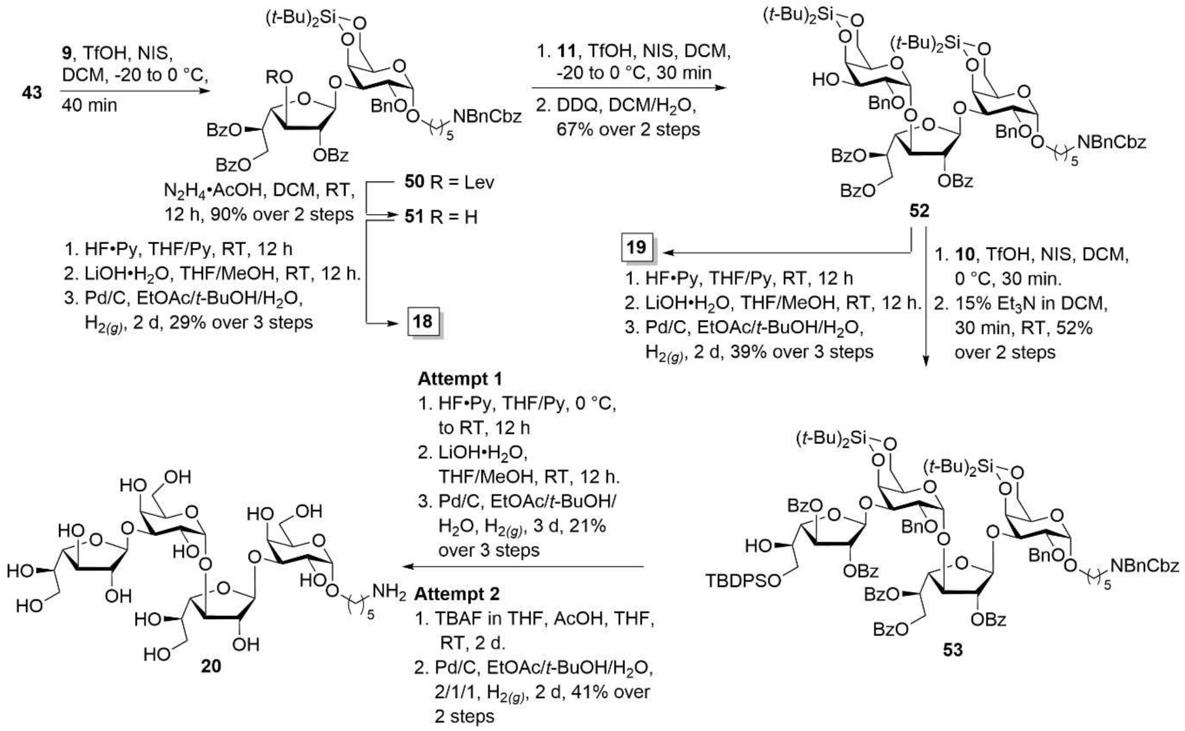


Figure 2. Structure of KP A) O1 antigen-related glycans **15**–**35** and B) O2ac antigen-related glycans **36**–**41**.

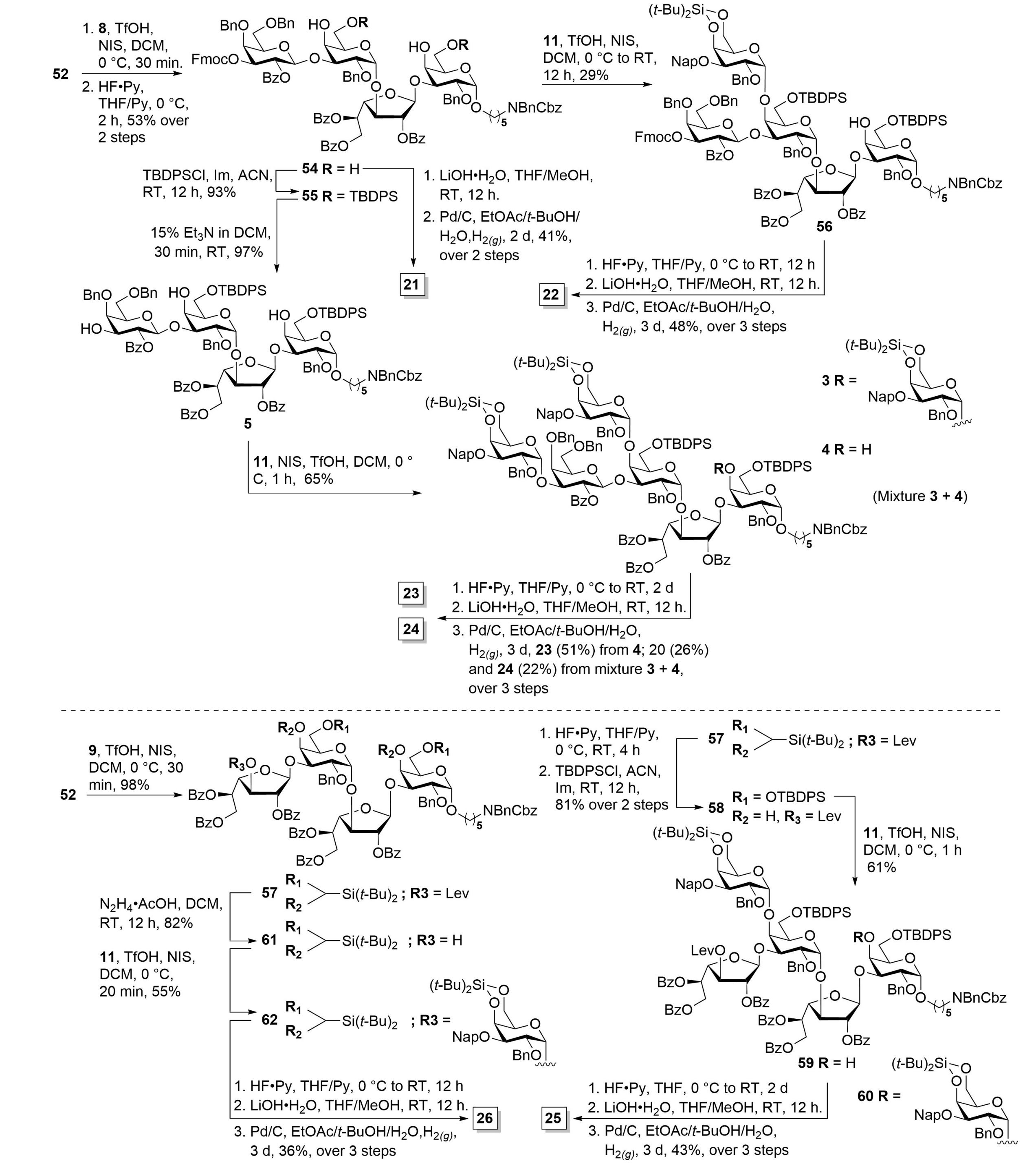
by glycosylation with donor **8** at 0 °C promoted by NIS/TfOH afforded pure hexasaccharide **49** in 41 % over two steps. Global deprotection of protected oligosaccharides **47**–**49** required three steps. Silyl ethers were cleaved with HF•Py[21] before hydrolysis of all remaining benzyl ether and benzyloxy carbamate groups with lithium hydroxide and Pd/C-catalyzed hydrogenolysis in a mixture of ethyl acetate, *t*-butanol and water afforded antigens **15** (33 %), **16** (35 %) and **17** (23 %) over three steps.

Oligosaccharides **18**–**20** resemble the repeating unit of the KP O2a antigen. Coupling glycosyl acceptor **43** with donor **9** using NIS/TfOH at −20 °C did not result in full conversion (Scheme 2), as indicated by thin-layer chromatography (TLC). However, warming the reaction to 0 °C for 30 minutes led to complete conversion to desired disaccharide **50**. Donor **9**, with more electron-withdrawing groups like benzoyl and levulinoyl esters, showed reduced reactivity. The use of a disarming ester group on the galactofuranose moiety might facilitate the construction of linear oligogalactofuranoses. However, arming with benzyl ethers has shown intrinsic steric and stereoelectronic effects, potentially causing deletion sequences during galactofuranose chain growth.[22] Levulinyl ester cleavage using hydrazine acetate furnished acceptor **51** in 90 % yield over two steps. Glycosylation of the resulting alcohol **51** with **11** followed by oxidative cleavage of NAP ether using DDQ afforded trisaccharide **52**. Extending the O2a antigen backbone by coupling **52** with donor **10** using successive amounts of the promoter TfOH/NIS (0.3/1.8 eq) furnished the protected tetrasaccharide, followed by Fmoc removal to give the desired tetrasaccharide **53** in 52 % yield over two steps. The moderate glycosylation efficiency is likely influenced not only by the disarmed nature of donor **10** but also by steric hindrance from the 4,6-di-tert-butylsilyl (DTBS) group on the galactosyl residue already installed at the O-1 position of the galactofuranosyl residue. This bulky DTBS protecting group may restrict access to the reactive site, further reducing the overall yield. Stepwise global deprotection of glycans **51** and **52** involved silyl ether cleavage, followed by ester hydrolysis and catalytic hydrogenation to afford desired antigens **18** and **19**. Global deprotection of **53** with HF•Py cleaved the silyl groups, followed by hydrolysis of esters using LiOH•H2O and hydrogenation using palladium on carbon (Pd/C) furnished **20** in 21 % yield over three steps. An attempt to remove the protecting groups of **53** using tetrabutylammonium fluoride (TBAF) buffered with acetic acid (AcOH) and Pd/C afforded antigen **20** in a better yield (41 %) along with the acetylated product (23 %). The AcOH-buffered TBAF system is used primarily for the deprotection of the tert-butyldiphenylsilyl (TBDPS) group, a sterically hindered protecting group. However, under these conditions, the high concentration of TBAF also contributes significantly to the removal of benzoyl groups. The exact mechanism likely involves nucleophilic attack by fluoride ions generated by TBAF on the carbonyl carbon of the benzoyl group, facilitating cleavage and promoting subsequent hydrolysis.

Synthetic glycans, resembling the KP O2a-O1 and O2afg-O1 repeating units, were prepared next (Scheme 3). Glycosylation of trisaccharide **52** with thioglycoside **8** followed by silyl ether cleavage afforded tetrasaccharide **54** that was converted to **21** by global deprotection. The regioselective TBDPS protection of the primary hydroxyl groups in **54** made the C4 hydroxyl group in the tetrasaccharide **55** better accessible. Placement of two *α*-glycosidic linkages by reaction of excess silylated thiogalactoside **11** (4 eq) with tetrasaccharide **55** furnished pentasaccharide **56** rather than the desired hexasaccharide. Additional promoter, higher reaction temperature (0 °C) and longer reaction times did not remedy the situation. Apparently, the C4 hydroxyl group in the reducing end galactose is sterically hindered by the presence of a C6 TBDPS ether protective group to engage as nucleophile in glycosylation reactions. The protecting groups of **56** were

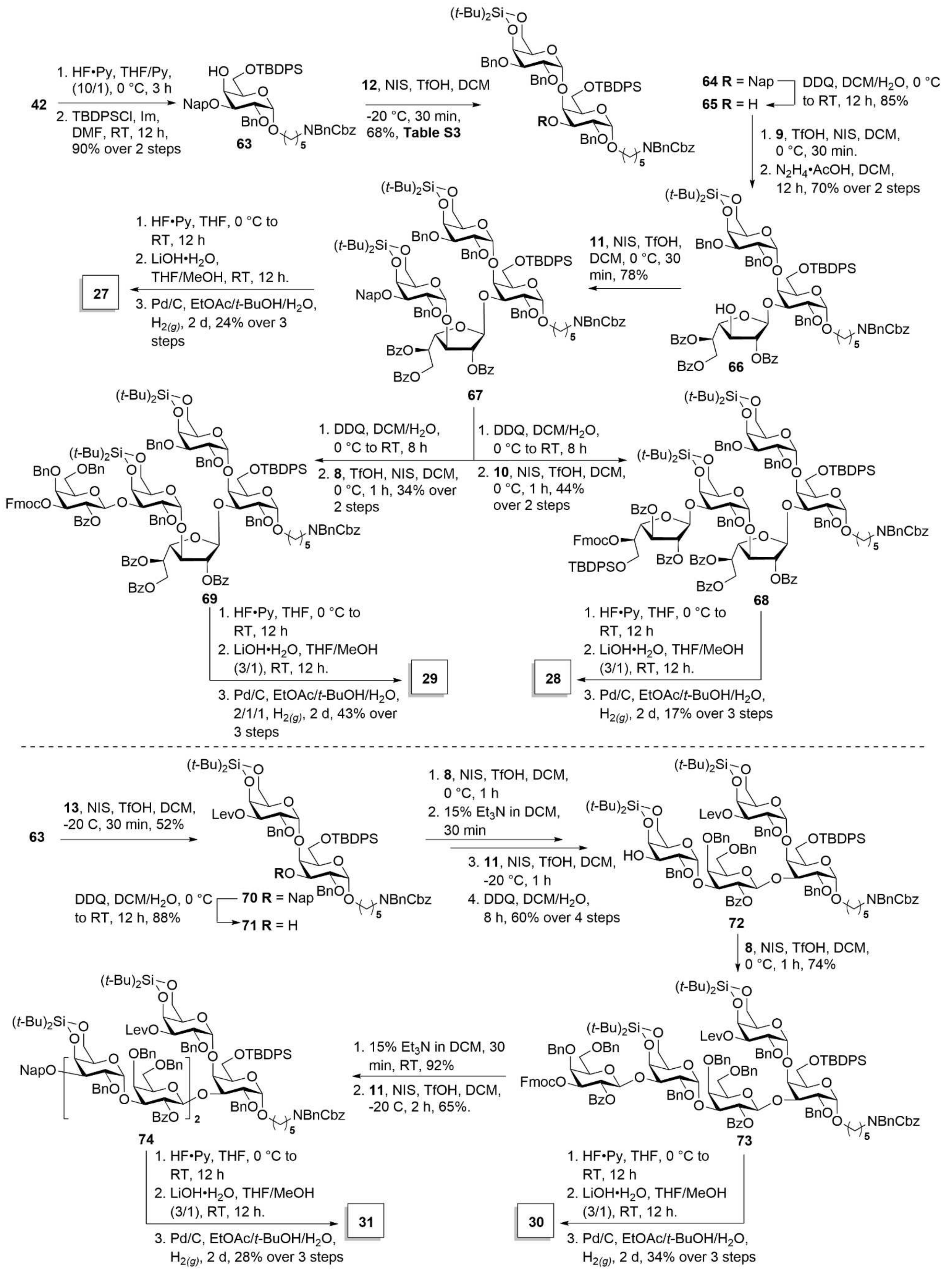


Scheme 2. Synthesis of O2a antigen-related oligosaccharides **18**–**20**.

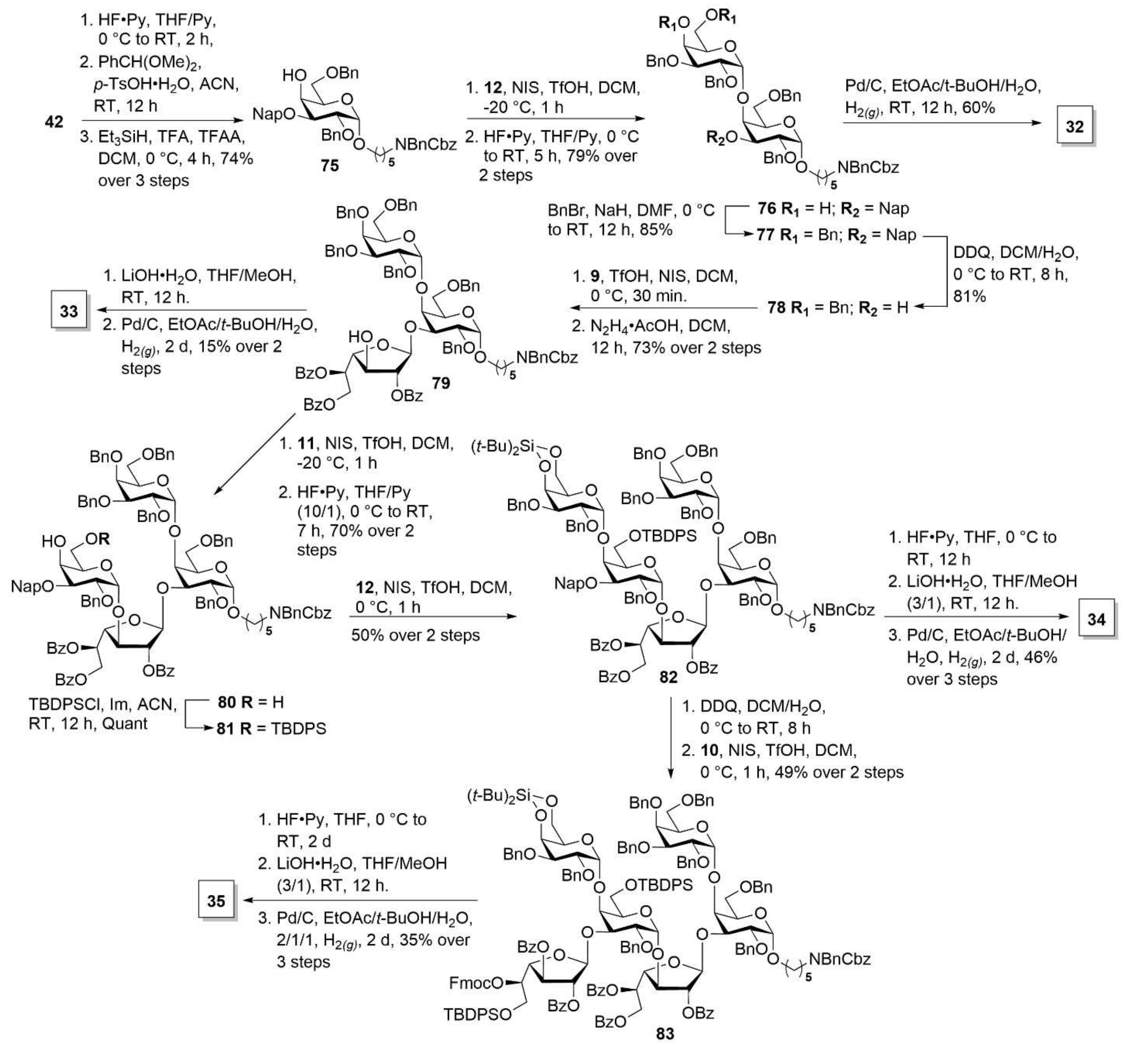


Scheme 3. Synthesis of oligosaccharides **21**–**26** related to O2a-O1, O2a, and O2afg-O1 antigens

removed in a three-step procedure and 48 % overall yield to furnish antigen **22**. Removal of the Fmoc group in **55** furnished tetrasaccharide triol **5**, ready for triple glycosylation with glycosyl donor **11**. We investigated reaction conditions suitable for the construction of heptasaccharide **3** (Table S1, Supporting Information). No conversion of **3** was observed when the reaction was activated with TMSOTf/NIS at −20 °C to 0 °C. Incomplete coupling of **4** (15 %) and partial recovery of acceptor **5** (40 %) were observed. Changing the Lewis acid from TMSOTf to TfOH did not improve the results. Glycosylation attempts with successive NIS addition with 1.0 equivalent of TfOH provided only trace amounts of the desired product. Increasing the amount of NIS to five equivalents resulted in a mixture of **3** and **4** that could not be separated by silica column chromatography, due to similar elution times. Subsequent three-step global deprotection of both **4** and the inseparable mixture of **3**–**4**, including silyl removal, ester hydrolysis, and benzyl ether hydrogenolysis



Scheme 4. Synthesis of oligosaccharides **27**–**31** related to the O2afg and O2afg-O1 antigens.

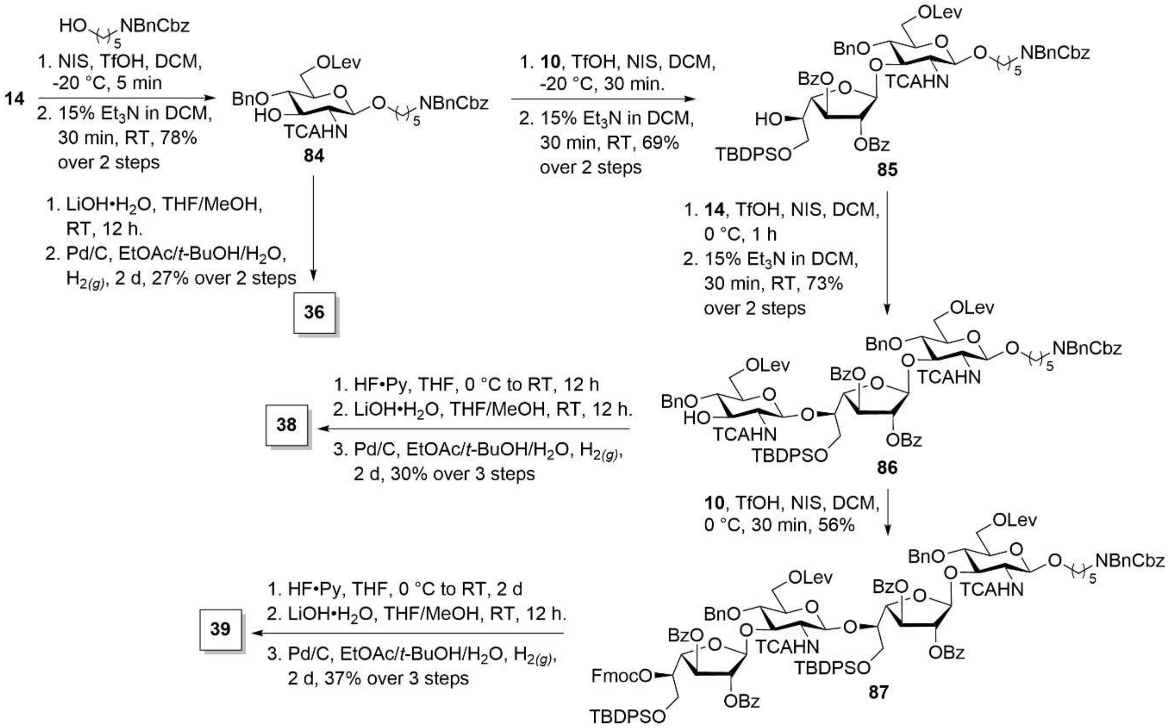


Scheme 5. Synthesis of oligosaccharides **32**–**35** related to O2afg-O1 antigen.

afforded desired antigens **23** and **24**. Heptasaccharide **24** was isolated as a pure compound after HPLC separation (See Supporting Information). Next, we synthesized pentasaccharides **25** and **26** using a linear glycosylation strategy (Scheme 3). Glycosylation of trisaccharide **52** with galactosylfuranose thioglycoside **9** afforded tetrasaccharide **57** in 98 % yield, before the protective groups were adjusted by *tert*-dibutylsilyl cleavage and regioselective protection of the primary hydroxyl group with a bulky silyl TBDPS ether to afford **58** in 81 % yield over two steps. The installation of galactose thioglycoside **11** onto tetrasaccharide **58** required more NIS promoter at 0 °C for 1 h to obtain 61 % pentasaccharide **59** but no di-glycosylated product **60**. Even with more promoter and higher reaction temperatures, pentasaccharide **60** was not obtained due to the bulky protecting groups near the nucleophile (Table S2, Supporting Information).

Pentasaccharide **59** was relieved of all protective groups by global deprotection with HF•Py, LiOH•H_2O and hydrogenolysis with Pd/C to afford pentasaccharide **25**, as a portion of the O2afg antigen. Pentasaccharide **26**, which covers another portion of the O2a antigen, was prepared by delevulinylation of **57** with hydrazine acetate to afford tetrasaccharide acceptor **61**. Galactosylation of **61** with **11** furnished 55 % of *α*-linked pentasaccharide **62** before silyl ether cleavage. Ester hydrolysis and hydrogenolysis afforded antigen **26** in 36 % yield over three steps after HPLC purification.

To overcome the steric hindrance of the bulky TBDPS protecting group during the galactosylation of oligosaccharide acceptors, the galactose branch was introduced prior to chain elongation. Silyl ether cleavage of **42**, followed by regioselective TBDPS protection of the 6-hydroxyl group yielded 90 % of differentially protected monosaccharide **63** (Scheme 4). *α*-



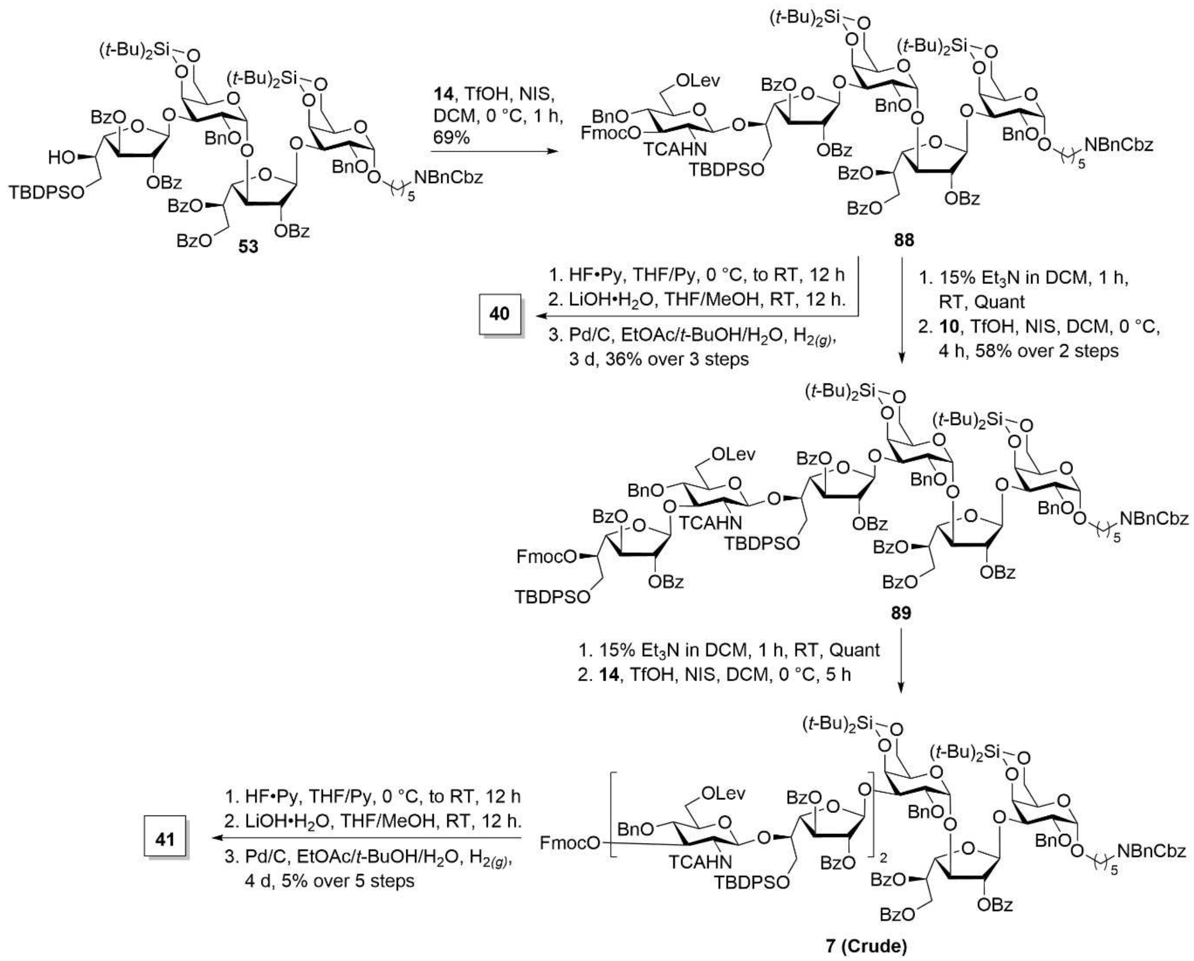
Scheme 6. Synthesis of monosaccharide **36**, trisaccharide **38** and tetrasaccharide **39** related to the O2a antigen.

Galactosylation of the C4 hydroxyl of **63** using **12**[19a] mediated by NIS/TfOH at -20 °C afforded disaccharide **64** (68 %). Additional NIS/TMOTf promoter and prolonged reaction times failed to improve the yield, possibly because the bulky C6 TBDPS in **63** sterically hinders the nucleophilic attack (Table S3, Supporting Information).[23] The 2-NAP ether in **64** was cleaved using DDQ in a DCM/water mixture to afford **65** in 85 % yield. Glycosylation with galactofuranose **9** and delevulinylation using hydrazine acetate afforded trisaccharide **66** in 70 %. α-Galactosylation with building block **11** furnished tetrasaccharide **67**. Cleavage of the 2-NAP ether using DDQ followed by glycosylation with either building block **8** or **10** afforded pentasaccharides **68** (44 % yield) and **69** (34 % yield). Oligosaccharides **67–69** were globally deprotected to afford tetrasaccharide antigen **27** (24 %), pentasaccharide **28** (17 %) and pentasaccharide **29** (43 % yield). Next, oligosaccharides **30** and **31** that resemble the O2afg-O1 antigen were prepared. Disaccharide **70** was obtained in 52 % yield by NIS/TfOH promoted condensation of **63** and thioglycoside **13**.

Oxidative cleavage of 2-NAP ether protection using DDQ generated disaccharide acceptor **71** that was coupled with **8**, before Fmoc cleavage and glycosylation with thioglycoside **11** delivered an α-linked tetrasaccharide. Cleavage of the 2-NAP ether afforded 60 % acceptor **72** over four steps. Glycosylation of fragments **8** and **72** using the NIS/TfOH system at 0 °C smoothly produced compound **73** in a 74 % yield. Fmoc removal and final glycosylation with **11** afforded hexasaccharide **74**. Three-step removal of silyl ether, esters and the permanent protecting groups afforded oligosaccharides **30** (34 %) and **31** (28 % yield).

Finally, we synthesized oligosaccharides **32–35** that resemble the O2afg-O1 antigen in either one or two galactose branches (Scheme 5). The silyl ether in **42** was cleaved with HF•Py, followed by the installation of a 4,6-*O*-benzylidene acetal and regioselective ring opening using a mixture of triethylsilane, trifluoroacetic acid and trifluoroacetic anhydride to afford monosaccharide **75** in 74 % yield over three steps. The glycosylation of acceptor **75** bearing a C6 benzyl ether protecting group with donor **12** provided a silylated disaccharide, followed by silyl ether cleavage afforded 79 % of the desired product **76**. A bulky protective group such as TBDPS at C6 strongly influences the reactivity of the C4-hydroxyl of galactose while the non-bulky protecting group such as benzyl ether resulted in a smooth reaction. Disaccharide **76** was deprotected via Pd/C-catalyzed hydrogenolysis in a mixture of EtOAc/*t*-BuOH/H_2O to furnish **32** in 60 % yield.

The installation of benzyl ether groups in **76** was the key to the synthesis of pentasaccharide **34** and hexasaccharide **35**. Removal of the silyl protecting group followed by treatment with benzyl bromide and sodium hydride (85 %) and oxidative cleavage of the 2-NAP ether afforded **78** (81 % yield). The union of **9** and **78** furnished fully protected trisaccharide **79** in 73 % over two steps. Compound **79** was treated with lithium hydroxide and Pd/C-catalyzed hydrogenolysis to give **33** in 15 % over two steps. Elongation of the tetrasaccharide chain was achieved using **11**, which bears a C3 2-NAP temporary group. After treatment with HF•Py, this process afforded tetrasaccharide **80** in 70 % yield over two steps.



Scheme 7. Synthesis of oligosaccharides **40** and **41** related to O2a-O2c antigen.

The regioselective placement of a non-bulky group at C6 in the presence of a free hydroxyl at C4 is challenging at the tetrasaccharide level. A TBDPS ether was placed at the C6 primary alcohol of diol **80** prior to using NIS/TfOH-mediated glycosylation with **12** at 0 °C, producing 50 % pentasaccharide **82** over two steps. Cleavage of the 2-NAP ether protecting group readied the pentasaccharide acceptor for glycosylation with **10** to obtain doubly branched hexasaccharide **83** (49 % over two steps). Global three-step deprotection furnished, after HPLC purification, antigens **34** (46 % yield) and **35** (35 % yield).

The construction of oligosaccharides **36–38** resembling the KP O2c antigen was performed via a linear synthesis using protected building blocks **10** and **14** (Scheme 6). The reaction of glucosamine **14** with the aminopentanol linker in the presence of TfOH/NIS at −20 °C, followed by Fmoc removal afforded 78 % monosaccharide **84** over two steps. Subsequent global deprotection afforded monosaccharide **36** in 27 % yield. Placement of galactofuranose **10** onto **84** by activation with NIS/TfOH afforded protected disaccharide that was treated with 15 % triethylamine in dichloromethane to cleave the temporary Fmoc protecting group for chain elongation. Galactofuranose building block **10** bearing ester groups performed better at 0 °C than at lower temperatures. Glycosylation of **85**, followed by Fmoc removal afforded trisaccharide **86** that was subjected to global deprotection to furnish trisaccharide **38** in 30 % yield over three steps. Trisaccharide **86** was extended by glycosylation with **10** to furnish tetrasaccharide **87** in 56 % yield, followed by deprotection with excess HF•Py and a longer reaction time was required to achieve complete cleavage of the bulky TBDPS ether before hydrolysis with LiOH•H_2O and hydrogenation yielded 37 % tetrasaccharide **39** over three steps.

Pentasaccharide **40** and heptasaccharide **41** were synthesized to cover the entire KP O2c antigen structure (Scheme 7). The above-obtained tetrasaccharide **53** was glycosylated with **14**, yielding fully protected pentasaccharide **88** followed by global deprotection delivered pure pentasaccharide **40** in 36 % yield over three steps. In anticipation of installing the galactofuranose building block **10** at C3, the Fmoc group in **88** was cleaved by triethylamine to give pentasaccharide acceptor that was glycosylated with donor **10** to get pentasaccharide **89** in 58 % over two steps. After deprotection of the Fmoc group in **89**, the free hydroxyl group was subsequently coupled using **14** to give heptasaccharide **7** as an inseparable mixture. During global deprotection, Pd/C catalyzed hydrogenation cleaved the glycosidic bonds to produce deletion sequences **19**, **37**, and **40** as confirmed by NMR spectroscopy (see Supporting Information). Still, sufficient amounts of well-defined oligosaccharide **41** were obtained for glycan array studies of sera from KP-infected patients.

Glycan arrays are useful tools to identify glycan epitopes and binding patterns of antibodies present in serum.[24] Synthetic glycans **15**–**41** and unrelated oligosaccharides were printed onto glass slides using methods described previously (Figure S2).[25] In total, 52 serum samples of patients infected with KP were tested for the presence of glycan-binding antibodies on a glycan array. Serum samples from ten healthy individuals vaccinated against Severe acute respiratory syndrome coronavirus type 2 (SARS-CoV2) and ten patients infected with SARS-CoV2 served as negative controls (for details see Supporting Information).

Antibody responses were observed across all tested immunoglobulin types (immunoglobulin G (IgG), immunoglobulin M (IgM), immunoglobulin A (IgA)) with generally higher antibody levels in the sera of KP-infected patients (Figure 3, Figure S3) compared to uninfected patients. In addition, patients showed elevated antibody levels against a rhamnose trisaccharide that served as positive control, because of the abundance of anti-rhamnose antibodies in human sera[26] and a stronger signal compared to rhamnose monomers. In the Klebsiella positive group, IgG-antibody titers against structures **17**, **20**, **26**, **28** and **30** were in particular elevated compared to the control group that was not infected with KP (Figure 3A). The differences in IgM antibody titers were less striking when comparing the sera of KP-infected persons with healthy controls (Figure S3). IgA- antibody titers against structures related to O1 (**15**, **16**, **17**, **30**), O1-O2a (**21**, **23**, **24**), O2a (**26**) and O2afg (**35**) were significantly elevated in the Klebsiella positive group compared to the Klebsiella negative group (Figure 3B). Especially high IgA

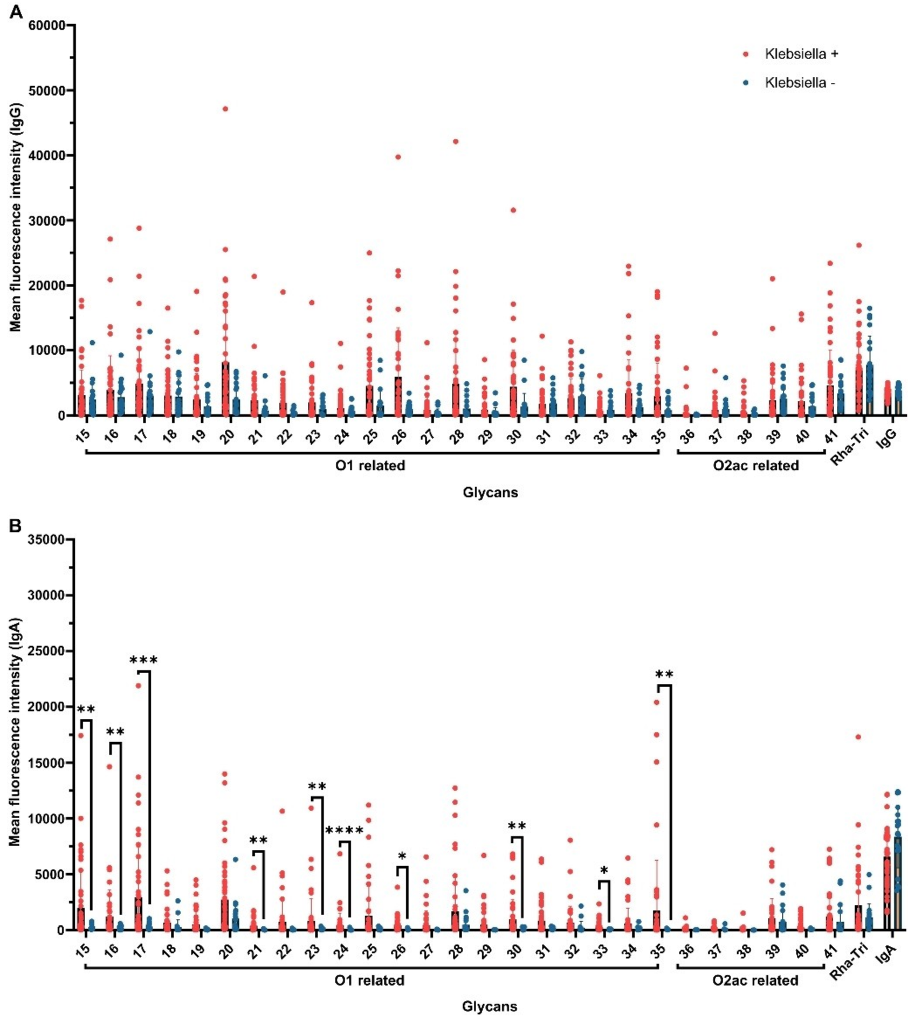


Figure 3. Glycan array analysis of sera from patients infected with KP as well as healthy controls. Antibody titers against synthetic glycans resembling the bacterial surface were evaluated A) IgG- antibody titers B) IgA- antibody titers. Klebsiella +: 47 samples of KP and CRKP-infected patients. Klebsiella −: healthy individuals vaccinated against SARS-CoV2 ($n=10$) and patients infected with SARS-CoV2 ($n=10$). IgG, IgA: isotype printing controls, Rha-Tri: rhamnose trimer as positive glycan printing control. Multiple Mann-Whitney test with multiple comparisons by False Discovery Rate (method by Benjamini, Krieger, and Yekutieli) set to 1%; *, $p<0.0332$; **, $p<0.0021$; ***, $p<0.0002$; and ****, $p<0.0001$; ns, not significant

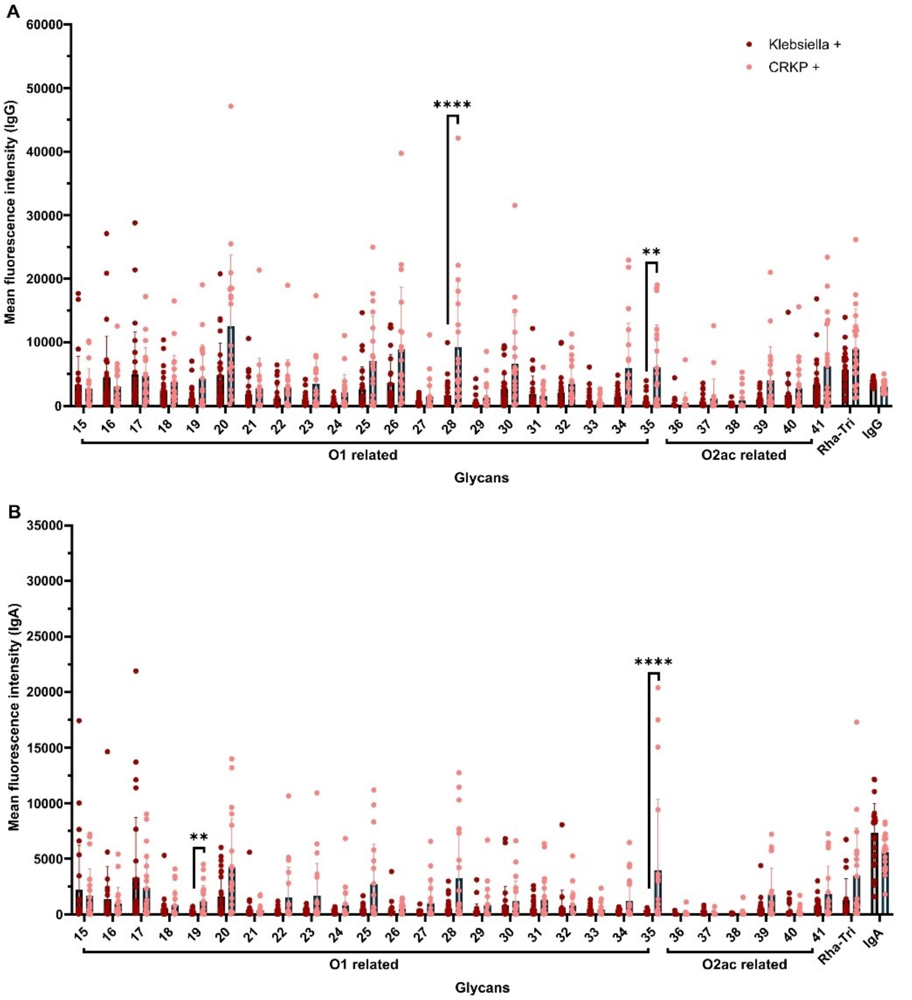


Figure 4. Glycan array analysis of sera from patients infected with KP or CRKP as well as healthy controls. Antibody titers against synthetic glycans resembling the bacterial surface were evaluated. Klebsiella + and CRKP + groups A) IgG- antibody titers B) IgA- antibody titers. Klebsiella +: patients infected with K. pneumoniae ($n=27$). CRKP +: patients infected with CRKP ($n=20$). IgG, IgA: isotype printing controls, Rha-Tri: rhamnose trimer as positive glycan printing control. Multiple Mann-Whitney test with multiple comparisons by False Discovery Rate (method by Benjamini, Krieger, and Yekutieli) set to 1%; $p<0.0332$; **, $p<0.0021$; ***, $p<0.0002$; and ****, $p<0.0001$; ns, not significant

levels in individual patients were observed for structures **15**, **16**, **17** and **35**. IgM and IgA antibody signals were generally lower than IgG possibly due to the time points and method of sampling. The blood samples were taken on average two weeks after the first positive culture, and possibly too late to detect significant IgM levels. During the primary immune response, IgM is the first class of antibodies that is produced, followed by a class switch to IgG or IgA.[27] IgA is mainly found on mucous membranes of the respiratory and digestive tract.

Different antibody types target distinct epitopes as IgA, IgM and IgG antibody patterns are differing. However, strong IgG binding to structures **17** (O1), **26** (O2a) and **30** (O1), as well as significantly increased IgA levels, was observed. Notably, these structures are composed exclusively of one O-antigen type and of interest for the development of vaccine candidates to achieve a broad immune response involving different immunoglobulin types.

IgA antibodies recognizing the O2afg repeating unit (**33**) were significantly higher in KP patients, while IgG levels for the same antigen did not differ. The sera of infected patients revealed elevated antibody levels for structures containing two repeating units of O1 (**15**) and O2a (**20**), while binding to structures composed of only one repeating unit did not show significantly elevated antibody production on the array. The tested O2c antigen minimal epitope, trisaccharide **38**, which consists of the repeating unit and an additional N-acetylglucosamine, was not recognized. Antibodies present in sera recognized at least two repeating units as present in tetrasaccharide **39**. Hence, the minimal epitopes include one repeating unit for serotype O2a or two repeating units for serotypes O1, O2a and O2c, respectively.

Glycans activate B-cells in a T-independent manner, multivalent binding and B-cell receptor cross-linking require larger molecules.[28] The most complex structures tested in this study are oligosaccharides **24** and **41**, yet, the immune responses are not higher than for shorter glycans. Short glycans consisting of two repeating units are sufficient for antibody recognition. Even a T-dependent formation of antibodies in response to bacterial infection and full LPS is conceivable and should be investigated. The IgG signal intensities were stronger for oligosaccharides containing several repeating units such as **17** and **20**. These longer structures are good vaccine leads.

Some immune response against KP-glycans is expected to be found in control samples, considering that KP is both a commensal and an opportunistic pathogen.[1] In addition, individuals may have been infected with KP in the past. Indeed, this could explain the stronger signals in healthy controls for certain glycans that were tested (e.g. **39**). Additionally, galactofuranose is found in bacteria, fungi and protozoa and a cross-reactivity of antibodies targeting other galactofuranose-containing structures is conceivable (e.g. **20**).[29]

Some patients with only very low antibody titers may have been infected with KP strains with other O-LPS types that were not tested on the array or have not developed a strong immune response. Sera from CRKP-infected patients contained significantly higher IgG-antibody titers against structures **28** and **35** (Figure 4A), while IgA- levels against **19** (O2a antigen with an additional galactose unit) and **35** (consisting of two repeating units of O2afg antigen) were significantly higher in CRKP patients than in KP patients (Figure 4B). IgM levels did not show significant differences between the groups (Figure S4). Trisaccharide **33**, containing one repeating unit of the O2afg antigen, did not show a high binding response. Galactofuranose containing O2 antigens in general and O2afg, in particular, have been linked to multidrug- resistant strains[30] and O2afg is the major antigen in the CRKP- associated sequence types ST258, ST307 and ST512.[31]

The immune response to CRKP infections is relatively poorly understood but recent advances show differences according to disease severity on a cellular level in single cell-RNA-sequencing experiments.[32] Higher immune responses in CRKP-infected patients could be a result of prolonged exposure to the pathogen and chronic progression of the disease. Strain O2 has previously shown a prevalence of 50 % in carbapenem- resistant strains and reduced immunogenicity. However, individual antibodies offer very effective protection against infections with strain O2 in in vivo mouse models.[29a] Our glycan array results suggest that certain antigens are better recognized by antibodies in the serum. Thereby, substructures can be identified that are best suited as vaccine candidates that offer protection against multi-drug-resistant strains.

Some KP patients showed elevated antibody levels against variations of the O1 antigen (**15**, **16** and **17**). O1, the predominant clinical isolate,[9c,30a] has been linked with high bacterial dissemination and organ colonization.[33] IgA antibodies prevent pathogen entry at mucosal surfaces and elevated secretory IgA (sIgA) levels are an indicator of pathogen control at mucosal sites.[34] Even though mucosal sIgA was not determined, IgA levels in sera from KP-infected patients were higher compared to non-infected individuals. The role of serum IgA has been linked with IgA-mediated intracellular killing via phagocytosis in vaccine studies for other bacteria species such as *Streptococcus pneumoniae*[35] and *Neisseria meningitides*.[36] The spike protein of SARS-CoV2 is highly N- and O-glycosylated[37] and contains galactose units that may be responsible for elevated levels of antibodies against galactose in patients and vaccinated controls.

A vaccine containing the lead structures for the O1 and O2 strains would cover 64 % of the strains globally, while glycans against O3 and O5 cover an additional 26 %.[33a] Oligosaccharides representing the O1 (**17**), O2a (**20**), O2afg (**35**) and O2ac antigens (**39**) were strongly recognized by IgG and IgA antibodies in the sera of patients infected with KP or antibiotic-resistant strains. Oligosaccharide antigens that combine different antigens such as O2a and O2afg (**28**) or O2a and O2ac (**41**) are attractive leads for vaccine development for broad serotype coverage. Glycan array monitoring of antibody responses in patient sera or other body fluids to KP antigens could provide an alternative to the currently used K and O serotyping.

Conclusion

The total synthesis of 27 oligosaccharides resembling various substructures of lipopolysaccharides from KP strains O1, O2a, O2afg and O2ac antigens served as a basis to identify promising glycan epitopes for vaccine development against KP infections. Glycan arrays containing these glycans were used to screen sera of patients infected with KP. Several antigens were strongly recognized by antibodies in the sera and have potential as vaccine leads to cover the strains O1 (**17**), O2a (**20**), O2afg (**35**) and O2ac (**39**) or even two strains at once such as O2a and O2afg (**28**) or O2a and O2ac (**41**). The evaluation of such semi-synthetic glycoconjugates as vaccine candidates in KP infection mouse models is ongoing.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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